

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Data engineer, social-media collection and measurement

Lahore, Pakistan · +44 7482 376417 · asad.ikram53@gmail.com · linkedin.com/in/asad-ikram98 · asad-ikram-mc.github.io/portfolio · web-scraping-guide.com

EDUCATION

2024 – 2025 awarded Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Credit-weighted average 70.5/100 (Distinction threshold 70). Masters project 78/100. Supervisor: Dr Ella Haig. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80).
2017 – 2021 awarded Apr 2022	BSc Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan. Four years full time, 130 credit hours completed. CGPA 2.75 on a 0.00–4.00 scale.

MASTER'S THESIS

Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies. Collected ~279,000 public posts on UK economic policy from X/Twitter, Reddit, YouTube and Quora via official APIs and licensed feeds, under ethics approval (Portsmouth TETHIC-2025-111094) and GDPR-compliant procedures. Fine-tuned a BERT classifier to macro-F1 0.878 against BiLSTM and lexicon baselines; human validation at Cohen's $\kappa \approx 0.78$; temperature-scaling calibration; McNemar and bootstrap significance testing; NRC emotion analysis linking anger and fear to negative policy reactions; LIME for interpretability; Prophet forecasting of the daily sentiment index (MAE ≈ 0.02). *The finding that shaped my research interest: the same policy event produced structurally different sentiment and emotion profiles on different platforms, which meant the cross-platform aggregate rested on an untested assumption, that the classifier measures the same construct everywhere it reads.*

RESEARCH OUTPUT

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Manuscript, complete; not published and not submitted.** Treats each social platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, purely from measurement differences between platforms.

No peer-reviewed publications to date.

PROFESSIONAL EXPERIENCE

Dec 2024 – present	Data Engineer, consultant, M+C Saatchi Fluency, United Kingdom (remote) - Design and operate the multi-platform social listening pipeline behind campaign intelligence for UK Government departments and for Amazon, Ford, Nike and Reckitt: Sprinklr and Brandwatch feeds plus custom collectors into AWS (ECS Fargate, Step Functions, Glue, Athena), with SageMaker NLP sentiment models. - Run a fleet of 68 collectors ingesting ten platforms daily into a governed 156-column master table, with an automated repair agent operating as a named CI workflow: when a source changes shape the system diagnoses the break, regenerates the extraction logic and tests it against live pages before shipping. 6.7 million items collected in a six-week window; 99%+ pipeline uptime. - Anti-bot handling, proxy rotation and TLS fingerprint management. Weekly KL-divergence drift alarms on pipeline output distributions. - Data governance on government-facing datasets: documented filter rules, permanent removal lists, pre-delete backups and weekly refresh cycles.
Feb 2025 – present	Founder and Chief Technology Officer, ArtemisAI Ltd (trading as ViralData), United Kingdom (remote) - Built the platform end to end: Meta Graph API ingestion of 30+ public pages daily into a layered lakehouse and Redshift warehouse; 111,000+ posts and 3.39 million comments processed. - Fine-tuned six encoder classifiers (sentiment, emotion, toxicity, topic, language and intent) from XLM-R, DistilBERT, DeBERTa-v3 and Detoxify bases on a ~20,000-item weighted dataset, with a per-sample loss weighted by label confidence, class balance and label source. Served from six SageMaker endpoints. - Built the labelling pipeline these models learn from: an LLM labels a stratified seed set with its reasoning, decision rules are mined from that reasoning into a prompt for a second, cheaper model, and low-confidence cases route back for review. Weekly retraining loop with three-tier confidence routing. - Multimodal analysis in production: CLIP image-text embeddings feeding a trained fusion classifier (macro-F1 0.93 on a hand-built gold set), a hybrid OCR pipeline (token-F1 0.928), and speech transcription. <i>Instrumented the measurement layer itself. The two annotation tiers agree on 82.9% of sentiment labels and 74.4% of emotion labels while the toxicity model reports 96.7% average confidence, and the dashboard's "accuracy" figure was inter-model agreement with no criterion behind it. That is what moved my interest from building these instruments to asking what their outputs mean.</i>
Feb 2022 – Aug 2024	Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan

- Led a three-engineer team owning 500+ regional collectors across 15 MENA countries on Kubernetes/EKS, gathering multilingual platform data in English, Arabic, Urdu and Hindi.
- Redesigned the layered data warehouse (RDL, ODL, ADL), improving ETL/ELT throughput by 60%. Promoted twice in two years.

2018 – present

Earlier and concurrent engagements

- **Fix.com**, Canada, consultant, May 2023 – present (part time): Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; Great Expectations validation with automated schema-drift detection, reducing bad production data by 70%.
- **VendueTech**, Ireland, Lead Data Engineer, Jun–Jul 2025: led three engineers building real-time auction ingestion; migrated Azure to AWS. **Prefe**, Italy, consultant, Sep–Dec 2024: rebuilt legacy collectors, cutting failure rate 60%. **CXG**, Dubai, Dec 2025 – Jan 2026: production collectors for luxury retail at 99%+ completeness against advanced anti-bot protections.
- **Freelance data engineer** (Level 2 seller), Dec 2018 – Feb 2026: 300+ five-star projects across 400+ websites for 50+ international clients.

METHODS AND TOOLS

Measurement	Generalizability theory, measurement invariance and differential item functioning, score linking, calibration; inter-coder reliability (Cohen's κ); McNemar and bootstrap testing; chi-square; Prophet forecasting.
NLP and ML	Transformer fine-tuning (BERT, XLM-R, DistilBERT, DeBERTa-v3); CLIP multimodal embeddings; Whisper transcription; OCR pipelines; BiLSTM; NRC and VADER lexicons; LIME and attribution.
Collection	Scrapy, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube); licensed feeds; proxy and anti-blocking management; TLS fingerprinting; detection anatomy; agentic browsers.
Engineering	Python, PySpark, SQL; AWS (Fargate, Glue, Athena, Redshift, SageMaker, Step Functions, Lambda, DynamoDB); Airflow; Docker and Kubernetes; PostgreSQL; CI/CD with automated quality gates. R functional rather than fluent.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25, the UK Foreign, Commonwealth and Development Office's flagship global scholarship, awarded to fund the MSc at Portsmouth.
Public work	Author and maintainer of web-scraping-guide.com , a free public engineering reference on web data collection covering libraries and tooling, anti-bot systems, detection anatomy, agentic browsers, legal and ethical constraints, production architectures and data quality.
Languages	Urdu (native), English (IELTS Academic 6.5 overall, March 2024; MSc taught and assessed in English). No Danish; I would begin learning on arrival.
Teaching	No formal university teaching. Mentored two junior engineers at Prefe and led three-engineer teams at Dubizzle Labs and VendueTech; wrote and maintain a public technical reference used by practitioners.